AMIT ROY

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Education

Mitra Institution (Bhowanipur Branch)
Secondary Education
2012 - 2020 | Kolkata

Barisha High School
Higher Secondary Education
2020 - 2022 | Kolkata

RCC Institute of Information Technology
Bachelor of Technology (Information Technology)
11/2022 - 06/2026 | Kolkata

Technical Skills

Programming Languages — Java, Javascript, Python, HTML, CSS

Libraries/Frameworks — NodeJS, ExpressJS, ReactJS, Tailwind CSS

Databases — MongoDB, MySQL

Academic and Extracurricular Achievements

- Secured copyright from the Copyright Office, Government of India for a Barcode Scanner Using Computer Vision for Electricity Meter Interpretation.
- Cleared Smart India Hackathon 24 Internal Round
- Smart Bengal Hackathon 2025 Finalists

Languages

- English
- Hindi
- Bengali

Professional Experience

Research and Development Intern at Hi-Tech Systems & Services Ltd.
07/2025 - 06/2026 | Kolkata

- Designed and developed the Torpedo Monitoring System web application for real-time monitoring and management.
- Designed and developed the Substation Monitoring System web application with an intuitive interface for monitoring and control.
- Developed RESTful API services to control Pan-Tilt (PTZ) camera movement in the Substation Monitoring System.
- Developed the LevelState Android application for communication with hardware devices such as ELS 300+ and EDLI, enabling secure device activation, configuration, and data exchange.
- Designed and developed the LevelState Admin Panel (Full Stack) to securely manage registered devices, monitor device activation status, and provide remote customer support.
- Designed and developed a continuous thermal monitoring system for dumpers (trucks) using Raspberry Pi and a thermal camera, along with an Android application for downloading and managing recorded videos.
- Developed the Android application for the Skyclimber Monitoring System, enabling monitoring and interaction with the hardware system.
- Configured Beckhoff PLC digital input/output (I/O) modules using TwinCAT software and integrated the PLC with the ASLD software for seamless industrial automation and communication.

Projects

Dashboard for Swacchta and LiFE
An AI-powered dashboard for monitoring cleanliness and LiFE practices in offices, accessible from the Divisional Office. It uses image processing to detect deviations from Swacchta and Green Growth standards, triggering alerts for prompt corrective action at the sub-office level. Tech stack includes Express.js, Node.js, Javascript, MongoDB, EJS, Python, YOLO V8

Mindful AI
A web-application designed to act as a self-mental well-being testing tool to help users who find it hard to talk to people and approach therapists to share their situation. Analyzes mood using face, voice and text with other features such as community chat, journal writing, relaxing music, breathing exercises. Tech Stack used: React.js, Node.js, Express.js, MongoDB

DIY Drone Development
Designed and built two functional drones. The first, based on Arduino Uno, achieved stable manual flight through custom code and sensor integration. The second drone used a SpeedyBee F405 V3 flight controller, configured and fine-tuned with iNav navigation software to enable features like GPS hold, altitude control, and waypoint navigation.

Barcode Scanner Using Computer Vision For Electricity Meter Interpretation
Built a computer vision-based system to automate electricity meter reading and prevent fraud. After detecting the barcode, the meter reading area is auto-cropped, stored in a database, and the reading is extracted using OCR. Utilized YOLOv8 for object detection and integrated it with a Node.js backend and MongoDB database. The project is officially copyrighted by the Copyright Office, Government of India.